Homework 2 solution

Gerard (Lezhi) Wu

I used Python for the numerical work in this submission. Each assigned exercise has its own script. For Exercise 3.5(c), `exercise35.py` contains the standard-form simplex implementation used in the calculation.

Exercise 2.8

Part (a): feasible region

The constraints can be rewritten as

$$0 \leq x_1 \leq 3, \qquad x_1 \leq x_2 \leq 6 - x_1.$$

The feasible region is the triangle with vertices

$$(0,0), \qquad (0,6), \qquad (3,3).$$

A labelled plot of this triangle is in `HW2_explorer.ipynb`.

Part (b): standard form and basic feasible solutions

Add nonnegative slack variables s_1, s_2, and s_3. The standard-form system is

$$\begin{aligned} x_1 + x_2 + s_1 &= 6, \\ x_1 - x_2 + s_2 &= 0, \\ x_1 + s_3 &= 3, \\ x_1, x_2, s_1, s_2, s_3 &\geq 0. \end{aligned}$$

A **basis** is a choice of three linearly independent columns, one for each equality. A **basic feasible solution** is the resulting nonnegative five-variable solution after setting all nonbasic variables equal to zero.

Enumerating the bases in Python gives:

Basis	$(x_1, x_2, s_1, s_2, s_3)$	**Extreme point** (x_1, x_2)
(x_1, x_2, s_1)	$(3, 3, 0, 0, 0)$	$(3, 3)$
(x_1, x_2, s_2)	$(3, 3, 0, 0, 0)$	$(3, 3)$
(x_1, x_2, s_3)	$(3, 3, 0, 0, 0)$	$(3, 3)$
(x_1, s_1, s_3)	$(0, 0, 6, 0, 3)$	$(0, 0)$
(x_2, s_1, s_3)	$(0, 0, 6, 0, 3)$	$(0, 0)$
(s_1, s_2, s_3)	$(0, 0, 6, 0, 3)$	$(0, 0)$
(x_2, s_2, s_3)	$(0, 6, 0, 6, 3)$	$(0, 6)$

Part (c): correspondence with extreme points

There is no one-to-one correspondence. A **degenerate** solution has at least one basic variable equal to zero, so several bases may describe the same extreme point.

Extreme point	Number of feasible bases	Reason
$(0, 0)$	3	$x_1 = x_2 = s_2 = 0$ at the point
$(0, 6)$	1	Only one feasible basis represents it
$(3, 3)$	3	$s_1 = s_2 = s_3 = 0$ at the point

Here, both $(0, 0)$ and $(3, 3)$ have multiple feasible bases, although each is one basic feasible solution vector.

Exercise 2.9

Part (a): Exercise 2.3(a)

Introduce slacks $s_1, s_2, s_3 \geq 0$:

$$\begin{aligned} -2x_1 + x_2 + s_1 &= 2, \\ -x_1 + x_2 + s_2 &= 3, \\ x_1 + s_3 &= 2. \end{aligned}$$

The objective is $z = -x_1 - 2x_2$. The feasible bases and objective values are:

Basis	$(x_1, x_2, s_1, s_2, s_3)$	z
(x_1, x_2, s_1)	$(2, 5, 1, 0, 0)$	-12
(x_1, x_2, s_3)	$(1, 4, 0, 0, 1)$	-9
(x_1, s_1, s_2)	$(2, 0, 6, 5, 0)$	-2
(x_2, s_2, s_3)	$(0, 2, 0, 1, 2)$	-4
(s_1, s_2, s_3)	$(0, 0, 2, 3, 2)$	0

The minimum is

$$x^* = (2, 5), \qquad z^* = -12.$$

Part (b): Exercise 2.3(b)

The first inequality is a greater-than constraint, so subtract a nonnegative surplus variable s_1; add slack s_2 to the second inequality:

$$\begin{aligned} x_1 - 2x_2 - s_1 &= 2, \\ x_1 + x_2 + s_2 &= 4, \\ x_1, x_2, s_1, s_2 &\geq 0. \end{aligned}$$

Enumerating the bases in Python gives:

Basis	(x_1, x_2, s_1, s_2)	$z = -x_1 - 2x_2$
(x_1, x_2)	$(10/3, 2/3, 0, 0)$	$-14/3$
(x_1, s_1)	$(4, 0, 2, 0)$	-4
(x_1, s_2)	$(2, 0, 0, 2)$	-2

Therefore

$$x^* = \left(\frac{10}{3}, \frac{2}{3}\right), \qquad z^* = -\frac{14}{3}.$$

More generally, if u and v are two optimal feasible solutions with common objective value z^* and $0 \le \lambda \le 1$, linear constraints imply that $w = \lambda u + (1-\lambda)v$ is feasible, while linearity of the objective gives

$$c^T w = \lambda c^T u + (1-\lambda)c^T v = \lambda z^* + (1-\lambda)z^* = z^*.$$

Hence w is also optimal and P^* is convex. In the two problems above, the optimum is unique, so $P^* = \{x^*\}$ in each case. If $u, v \in P^*$, then $u = v = x^*$, and

$$\lambda u + (1-\lambda)v = \lambda x^* + (1-\lambda)x^* = x^* \in P^*.$$

Exercise 2.11

Let a_i^T denote row i of A. If P is empty, the problem has no feasible point and therefore has no optimal solution. Suppose instead that P is nonempty and take any $x \in P$. Because the inequalities are strict, each slack is positive:

$$\delta_i = b_i - a_i^T x > 0.$$

Move in direction c by defining

$$y = x + \epsilon c, \qquad \epsilon > 0.$$

For every row with $a_i^T c > 0$, choose ϵ small enough that

$$\epsilon < \frac{\delta_i}{a_i^T c}.$$

Rows with $a_i^T c \le 0$ remain strict for every $\epsilon > 0$ because the move does not increase their left-hand side. Since there are finitely many rows, a positive ϵ satisfying all required upper bounds exists. Hence $Ay < b$, so $y \in P$.

The objective strictly improves:

$$\begin{aligned} c^T y - c^T x &= c^T(x + \epsilon c) - c^T x \\ &= \epsilon c^T c \\ &= \epsilon \|c\|_2^2 \\ &> 0, \end{aligned}$$

where the final inequality follows from $\epsilon > 0$ and $c \neq 0$. Every feasible x therefore admits a feasible point with a larger objective value, so no feasible point is optimal.

Exercise 3.4

Part (a): feasible set

Add slack variables $s_1, s_2 \ge 0$:

$$\begin{aligned} -x_1 + 3x_2 + s_1 &= 2, \\ -3x_1 + 2x_2 + s_2 &= 1, \\ x_1, x_2, s_1, s_2 &\ge 0. \end{aligned}$$

In the original coordinates the constraints give upper bounds on x_2:

$$0 \le x_2 \le \min\left(\frac{2 + x_1}{3}, \frac{1 + 3x_1}{2}\right), \qquad x_1 \ge 0.$$

The region is unbounded to the right. The notebook plots the portion with $0 \le x_1 \le 8$ and marks the continuation beyond the plotted window.

Parts (b) and (c): simplex method and unboundedness

Use the initial basis $B = (s_1, s_2)$. A **reduced cost** is the objective-rate change associated with increasing a nonbasic variable while preserving the equalities. For a minimization problem, a negative reduced cost permits an improving pivot.

The variable vector below is ordered as (x_1, x_2, s_1, s_2). If d is a simplex direction, a move of length $\theta \geq 0$ has the form $x_{\text{new}} = x + \theta d$.

Iteration	**Basis**	**Basic point** (x_1, x_2, s_1, s_2)	**Objective**	**Entering**	**Leaving**	**Ste**
0	(s_1, s_2)	$(0, 0, 2, 1)$	0	x_2	s_2	$1/$
1	(s_1, x_2)	$(0, 1/2, 1/2, 0)$	$-5/2$	x_1	s_1	$1/$
2	(x_1, x_2)	$(1/7, 5/7, 0, 0)$	$-27/7$	s_2	none	unbounde

The corresponding reduced costs and directions are:

Iteration	**Nonbasic reduced costs**	**Improving direction** d	**Ratio test**
0	$(r_{x_1}, r_{x_2}) = (-2, -5)$	$(0, 1, -3, -2)$	$\min(2/3, 1/2) = 1/2$, so s_2 leaves
1	$(r_{x_1}, r_{s_2}) = (-19/2, 5/2)$	$(1, 3/2, -7/2, 0)$	$(1/2)/(7/2) = 1/7$, so s_1 leaves
2	$(r_{s_1}, r_{s_2}) = (19/7, -11/7)$	$(3/7, 1/7, 0, 1)$	No component restricts an increase in s_2

At iteration 2, increasing s_2 gives the full standard-form direction

$$d = \left(\frac{3}{7}, \frac{1}{7}, 0, 1\right).$$

It satisfies $Ad = 0$ and $d \geq 0$, so every point

$$x(t) = \left(\frac{1}{7}, \frac{5}{7}, 0, 0\right) + td, \qquad t \geq 0,$$

remains feasible. Its objective decreases by

$$c^T d = -2\left(\frac{3}{7}\right) - 5\left(\frac{1}{7}\right) = -\frac{11}{7} < 0$$

per unit of t. The minimization problem is unbounded below. In the original two variables a simpler feasible ray is $(x_1, x_2) = (t, 0)$ for $t \geq 0$, whose objective is $-2t \to -\infty$.

Exercise 3.5

Part (a): simplex method

Add slack variables $s_1, s_2, s_3 \geq 0$:

$$\begin{aligned} x_1 - 2x_2 + s_1 &= 2, \\ x_1 - x_2 + s_2 &= 3, \\ x_2 + s_3 &= 3. \end{aligned}$$

The objective coefficient vector, equality matrix, right-hand side, and initial slack basis used by the Python implementation are

$$c = (0, -1, 0, 0, 0)^T,$$

$$A_{eq} = \begin{bmatrix} 1 & -2 & 1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}, \qquad b_{eq} = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}, \qquad B_{\text{set}} = (s_1, s_2, s_3).$$

The variable vector is ordered as $(x_1, x_2, s_1, s_2, s_3)$.

Iteration	Basis	Basic point $(x_1, x_2, s_1, s_2, s_3)$	Objective	Entering	Leaving	St
0	(s_1, s_2, s_3)	$(0, 0, 2, 3, 3)$	0	x_2	s_3	

Iteration	Basis	Basic point $(x_1, x_2, s_1, s_2, s_3)$	Objective	Entering	Leaving	St
1	(s_1, s_2, x_2)	$(0, 3, 8, 6, 0)$	-3	none	none	opti

At iteration 0, the nonbasic reduced costs are $(r_{x_1}, r_{x_2}) = (0, -1)$. Thus x_2 enters. Its simplex direction is

$$d = (0, 1, 2, 1, -1),$$

and only s_3 decreases, giving the ratio $3/1 = 3$ and the pivot shown in the table. At iteration 1, the nonbasic reduced costs are $(r_{x_1}, r_{s_3}) = (0, 1)$. There is no negative reduced cost, so the point is optimal. The zero reduced cost for x_1 also signals alternate optima: increasing x_1 uses direction

$$d_{\text{alt}} = (1, 0, -1, -1, 0).$$

The maximum feasible step is $\min(8/1, 6/1) = 6$, which reaches the second optimal endpoint $(6, 3)$.

One optimal basic feasible solution is

$$(x_1, x_2) = (0, 3), \qquad z^* = -3.$$

Part (b): all optimal solutions

The objective is $-x_2$, so its minimum is obtained by taking x_2 as large as possible. The third constraint gives $x_2 \leq 3$; therefore every optimum has $x_2 = 3$. Substitution into the first two constraints gives

$$x_1 \leq 8, \qquad x_1 \leq 6, \qquad x_1 \geq 0.$$

The second upper bound binds first. Hence the complete optimal set is

$$P^* = \{(x_1, x_2) = (t, 3) \mid 0 \leq t \leq 6\}.$$

The optimum is not unique: both $(0, 3)$ and $(6, 3)$ have objective value -3, as does every point on the line segment joining them.

Part (c): Python simplex execution

The computation is performed by importing the exercise solver:

```
from exercise35 import solve_exercise_35

solution = solve_exercise_35()
result = solution.simplex_result
```

The file `exercise35.py` performs this standard-form calculation in Python using `objective_coefficients`, `equality_matrix`, `right_hand_side`, and `initial_basis`. It starts from the same three slack variables, performs one pivot, and returns the result

$$(x_1, x_2, s_1, s_2, s_3) = (0, 3, 8, 6, 0), \qquad z^* = -3.$$